2

Experimental Design

2.1 Introduction

Although all chemists acknowledge the need to be able to design laboratory-based experiments, formal statistical (or chemometric) rules are rarely developed as part of mainstream chemistry. In contrast, a biologist or a psychologist will often spend weeks in carefully constructing a formal statistical design before investing months or years in time-consuming and often unrepeatable experiments and surveys. The simplest of experiments in chemistry are relatively quick and can be repeated, if necessary, under slightly different conditions; hence, not all chemists observe the need for formalised experimental design early in their career. For example, there is little point in spending a week for constructing a set of experiments that take a few hours to perform. This lack of expertise in formal design permeates all levels from management to professors and students. However, in contrast, some real-world experiments are expensive; for example, optimising conditions for a synthesis, testing compounds in a quantitative structure–activity relationships (QSAR) study or improving the chromatographic separation of isomers, and can take days or months of people's time, and it is essential to, under such circumstances, to have a good appreciation of the fundamentals of design.

There are several key reasons why the chemist can be more productive if he or she understands the basis of design, including the following four main areas.

- *Screening.* These types of experiments involve considering factors that are important for the success of a process. An example may be the study of a chemical reaction, dependent on the proportion of the solvent, catalyst concentration, temperature, pH, stirring rate and so on. Typically, 10 or more factors might be relevant. Which can be eliminated, and which should be studied in detail? Approaches such as factorial and Plackett–Burman designs (Sections 2.3.1–2.3.3) are useful in this context.
- *Optimisation.* This is one of the commonest applications in chemistry. How to improve a synthetic yield or a chromatographic separation? Systematic methods can result in a better optimum, found more rapidly. Simplex is a classical method for optimisation (Section 2.6), although several designs such as mixture designs (Section 2.5) and central composite designs (Section 2.4) can also be employed to find optima.
- *Saving time.* In industry, this is possibly the major motivation for experimental design. There are obvious examples in optimisation and screening, but even more radical cases, as in the area of quantitative structure–property relationships. From structural data of existing molecules, it is possible to predict a small number of compounds for further testing, representative of a larger set of molecules. This allows saving of enormous time. Fractional factorial, Taguchi and Plackett–Burman designs (Sections 2.3.2 and 2.3.3) are good examples, although almost all experimental designs have this aspect in mind.
- *Quantitative modelling.* Almost all experiments, ranging from simple linear calibration in analytical chemistry to complex physical processes, where a series of observations are required to obtain a mathematical model of the system, benefit from good experimental design. Many such designs are based around the central composite design (Section 2.4), although calibration designs (Section 2.3.4) are also useful.

An example of where systematic experimental design is valuable is the optimisation of the yield of a reaction as a function of reagent concentration and pH. A representation is given in Figure 2.1. In reality, this relationship is unknown in advance, but the experimenter wishes to determine that the pH and concentration (in mM) provide the best reaction conditions. Within 0.2 of a pH and concentration unit, this optimum happens to be pH 4.4 and 1.0 mM. Many experimentalists will start by guessing one of the factors, say concentration, and then finding the best pH at that concentration.

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For introductory purposes, multiple linear regression (MLR) analysis is used to relate the experimental response to the values of the factors, as is common to most texts in this area, but it is important to realise that other regression methods such as partial least squares (PLS) are applicable in many cases, as discussed in Chapter 6. Certain designs such as those discussed in Section 2.3.4 have direct relevance to multivariate calibration. In some cases, multivariate methods such as PLS can be modified by inclusion of squared and interaction terms as described below for MLR. It is important to remember, however, that in many areas of chemistry, quite a lot of information is available about a data set, and conceptually simple approaches based on MLR are often adequate.

2.2 Basic Principles

2.2.1 Degrees of Freedom

Fundamental to the understanding of experimental designs is the idea of degrees of freedom. An important outcome of many experiments is the measurement of errors. This can tell us how confidently a phenomenon can be predicted; for example, are we really sure that we can estimate the activity of an unknown compound from its molecular descriptors, or are we happy with the accuracy with which a concentration can be determined using spectroscopy? In addition, what is the weak link in a series of experiments? Is it the performance of a spectrometer or the quality of the volumetric flasks? Each experiment involves making a series of observations, which allow us to try to answer some of these questions, the number of degrees of freedom relating to the amount of information available for each answer. Of course, the more the degrees of freedom, the more certain we can be of our answers, but the more the effort and work required. If we have only a limited time, it is important to provide some information to allow us to answer all the desired questions.

Most experiments result in some sort of *model*, which is a mathematical way of relating an experimental *response* to the value or state of a number of *factors*. An example of a response is the yield of a synthetic reaction; the factors may be pH, temperature and catalyst concentration. An experimenter wishes to run a reaction under a given set of conditions and predict the yield. How many experiments should be performed in order to provide confident predictions of the yield at any combination of the three factors? 5, 10 or 20? Usually, the more the experiments, the more certain the predictions, but the greater the time, effort and expense. Hence, there is a balance, and experimental design helps to guide the chemist as to how many and what type of experiments should be performed.

Consider a linear calibration experiment, for example, measuring the peak height using electronic absorption spectroscopy as a function of concentration, at five different concentrations, illustrated in Figure 2.5. A chemist may wish to fit a straight-line model to the experiment of the form

$$y = b_0 + b_1 x$$

where y is the response (in this case the peak height), x is the value of the factor (in this case concentration) and b_0 and b_1 are the *coefficients* of the model. There are two coefficients in this equation, but five experiments have been

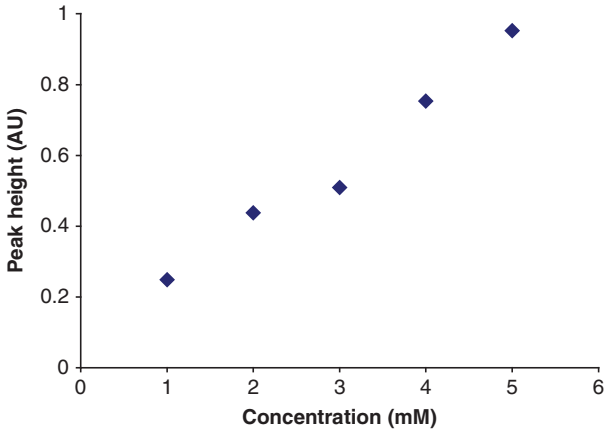


Figure 2.5 Graph of spectroscopic peak height against concentration at five concentrations.

Table 2.1 Three experimental designs.

	Design 1		Design 2		Design 3	
Experiment number	[A]	[B]	[A]	[B]	[A]	[B]
1	1	1	1	2	1	3
2	2	1	2	1	1	1
3	3	1	2	2	3	3
4	1	2	2	3	3	1
5	2	2	3	2	1	3
6	3	2	2	2	1	1
7	1	3	2	2	3	3
8	2	3	2	2	3	1
9	3	3				

experimental designs in Table 2.1. The aim is to produce a linear model of the form

$$y = b_0 + b_1 x_1 + b_2 x_2$$

The response y may represent the absorbance in a spectrum, the two x's the concentrations of two compounds. The value of P is equal to 3 in all cases.

- *Design 1.* This has a value of R equal to 0 and D of 6. There is no information about experimental error and all effort has taken into determining the model. If the relationship between the response and concentration is known with certainty (or this information is not of interest), this experiment may be a good one, but otherwise too little effort is taken in measuring replicates. Although this design may appear to provide an even distribution over the experimental domain, the lack of replication could, in some cases, lose crucial information.
- *Design 2.* This has a value of R equal to 3 and D of 2. There is a reasonable balance between taking replicates and examining the model. If nothing much is known about the certainty of the system, this is a good design taking into account the need to economise on experiments.
- *Design 3.* This has a value of R equal to 4 and D of 1. The number of replicates is rather large compared with the number of unique experiments. However, if the main aim is simply to investigate experimental reproducibility over a range of concentrations, this approach might be useful.

It is always possible to break down a set of planned experiments in this manner, and it is recommended as a first step before modelling.

1. An experiment is performed to study the relationship of the yield of a reaction as a function of pH, all other conditions are kept constant. Five pH values are studied, seven experiments are performed, three of which are at pH 6. The number of replicates is
   (a) 2
   (b) 3
   (c) 4
2. The number of degrees of freedom for estimating the lack-of-fit is always more than the number of coefficients in the model.
   (a) True
   (b) False

2.2.2 Analysis of Variance

A key aim of modelling is to ask how significant a factor is. In Section 2.2.1, we discussed about designing an experiment that allows sufficient degrees of freedom to determine the significance of a given factor; below we will introduce an important way of providing numerical information about this significance.

Most statistical packages produce ANOVA tables if required, and it is not always necessary to determine these errors manually, although it is important to appreciate the principles behind such calculations. However, for simple examples, a manual calculation is often quite quick and a good alternative to the interpretation of the output of complex statistical packages.

The use of ANOVA is widespread and is based on these simple ideas. Normally, two mean errors are compared, for example, one due to replication and the other due to lack-of-fit, although any two errors or variances may be compared. As an example, if there are 10 possible factors that might have an influence over the yield in a synthetic reaction, try modelling the reaction removing one factor at a time and see how much the lack-of-fit error increases. If not much relative to the replicates, the factor is probably not significant. It is important to recognise that reproducibility of the reaction has an influence over apparent significance also, as does the accuracy of measurement of the yield. If there is a large replicate error, then some significant factors might be missed out.

1. The sum of squares for predicted data is always less than the sum of squares for the observed data.
   (a) True
   (b) False
2. The mean sum of square lack-of-fit error is always less than the mean sum of square replicate error.
   (a) True
   (b) False
3. Nine experiments are performed as follows: condition 1 (1 experiment), condition 2 (2 experiments), condition 3 (3 experiments), condition 4 (2 experiments) and condition 5 (1 experiment). The total sum of square replicate error is 0.736. The mean square replicate error is
   (a) 0.105
   (b) 0.184
   (c) 0.245

2.2.3 Design Matrices and Modelling

The design matrix is a key concept. A design may consist of a series of experiments performed under different conditions, for example, a reaction at differing pHs, temperatures and concentrations. Table 2.6 illustrates a typical experimental set-up, together with an experimental response, for example, the rate constant of a reaction. Note the replicates in the final five experiments: in Section 2.4, we will discuss such an experimental design commonly called a *central composite* design.

2.2.3.1 Models

It is normal to describe experimental data by forming a mathematical relationship between the factors or independent variables such as temperature and a response or dependent variable such as a synthetic yield, a reaction time or a percentage impurity. A typical equation involving three factors might be of the form

$\hat{y} =$	(response)
$b_0 +$	(an intercept or average)
$b_1x_1 + b_2x_2 + b_3x_3 +$	(linear terms depending on each of the three factors)
$b_{11}{x_1}^2 + b_{22}{x_2}^2 + b_{33}{x_3}^2 +$	(quadratic terms depending on each of the three factors)
$b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3$	(interaction terms between the factors).

Notice the 'hat' on top of y, this is because the model estimates its value and is unlikely to give an exact value that agrees with the observed value because of error.

The explanation for these terms is as follows:

- The intercept is the estimated response when the values of all factors is 0: if the factors are coded to be centred at 0 (see Section 2.2.4.1), it is the estimated response in the centre of the design usually corresponding to the average values of the factors. It is an important term because a response of 0 is not normally achieved when the factors are at their average values. Only in certain circumstances (e.g. spectroscopy: if it is known, there are no baseline problems or interferences), this term can be safely ignored.

Table 2.6 Typical experimental design.

pH	Temperature (°C)	Concentration (mM)	Response (y)
6	60	4	34.841
6	60	2	16.567
6	20	4	45.396
6	20	2	27.939
4	60	4	19.825
4	60	2	1.444
4	20	4	37.673
4	20	2	23.131
6	40	3	23.088
4	40	3	12.325
5	60	3	16.461
5	20	3	33.489
5	40	4	26.189
5	40	2	8.337
5	40	3	19.192
5	40	3	16.579
5	40	3	17.794
5	40	3	16.650
5	40	3	16.799
5	40	3	16.635

- The linear terms allow for a direct relationship between the response and a given factor. For some experimental data, there are only linear terms. If pH increases, does the yield increase or decrease, and if so by how much?
- In many situations, quadratic terms are important. This allows curvature and is one way of obtaining a maximum or minimum. Most chemical reactions have an optimum performance at a particular pH, for example. Almost all enzymic reactions work in this way. Quadratic terms balance out the linear terms.
- In Section 2.1, we discussed the need for interaction terms. These arise because the influence of two factors on the response is rarely independent. For example, the optimum pH at one temperature may differ from that at a different temperature.

Some of these terms may not be very significant or relevant, but it is up to the experimenter to check using approaches such as ANOVA (Section 2.2.2) and the related significance tests (Section 2.2.4). In advance of modelling, it is often hard to predict which factors are important.

1. If we are interested in the relationship between a response y and two factors x_1 and x_2 and obtain a model with all possible intercept, linear, quadratic and interaction terms, how many terms are there in the model?

   (a) 4
   (b) 5
   (c) 6

2.2.3.2 Matrices

There are 10 terms or parameters in the above equation. Many chemometricians find it convenient to work using matrices. Although a significant proportion of traditional statistical texts often shy away from matrix-based notation, with modern computer packages and spreadsheets, it is easy and rational to employ matrices. In later chapters, we will use matrix notation more frequently. The design matrix is simply one in which

- the *rows* refer to experiments and
- the *columns* refer to individual terms in the mathematical model or equation linking the response to the values of the individual factors.

Table 2.7 Design matrix for the experiment in Table 2.6 using the model discussed in Section 2.2.3.1.

Intercept	Linear terms			Quadratic terms			Interaction terms		
b_0 Intercept	b_1 pH	b_2 Temp.	b_3 Conc.	b_{11} pH^2	b_{22} $Temp.^2$	b_{33} $Conc.^2$	b_{12} pH × temp.	b_{13} pH × conc.	b_{23} Temp. × conc.
1	6	60	4	36	3600	16	360	24	240
1	6	60	2	36	3600	4	360	12	120
1	6	20	4	36	400	16	120	24	80
1	6	20	2	36	400	4	120	12	40
1	4	60	4	16	3600	16	240	16	240
1	4	60	2	16	3600	4	240	8	120
1	4	20	4	16	400	16	80	16	80
1	4	20	2	16	400	4	80	8	40
1	6	40	3	36	1600	9	240	18	120
1	4	40	3	16	1600	9	160	12	120
1	5	60	3	25	3600	9	300	15	180
1	5	20	3	25	400	9	100	15	60
1	5	40	4	25	1600	16	200	20	160
1	5	40	2	25	1600	4	200	10	80
1	5	40	3	25	1600	9	200	15	120
1	5	40	3	25	1600	9	200	15	120
1	5	40	3	25	1600	9	200	15	120
1	5	40	3	25	1600	9	200	15	120
1	5	40	3	25	1600	9	200	15	120
1	5	40	3	25	1600	9	200	15	120

2.2.3.3 Determining the Model

The relationship between the response, the coefficients and the experimental conditions can be expressed in matrix form by

$$\hat{\boldsymbol{y}} = \boldsymbol{D}\,\boldsymbol{b}$$

as illustrated in Figure 2.12. Readers unfamiliar with matrix notation should read Section A.1.

It is simple to show that this is the matrix equivalent to the equation introduced in Section 2.2.3.1. It is surprisingly easy to calculate $\boldsymbol{b}$ (or the coefficients in the model) knowing $\boldsymbol{D}$ and $\boldsymbol{y}$ using MLR. This approach will be discussed in greater detail in Chapter 6.

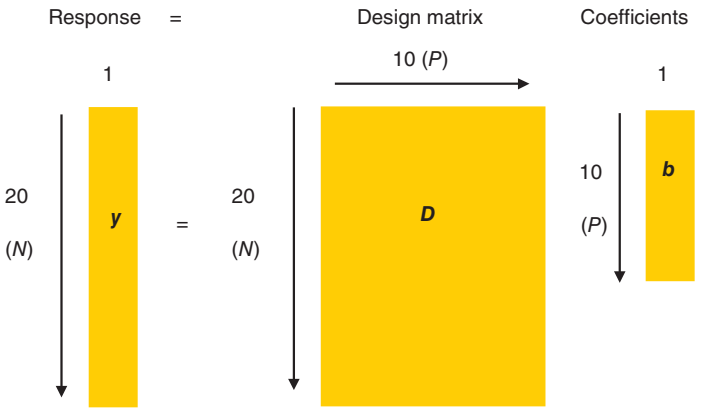


Figure 2.12 Relationship between response, design matrix and coefficients.

Table 2.8 The vectors $\boldsymbol{b}$ and $\hat{\mathbf{y}}$ for data in Table 2.6.

Parameters		Predicted y
b_0	58.807	35.106
b_1	−6.092	15.938
b_2	−2.603	45.238
b_3	4.808	28.399
b_{11}	0.598	19.315
b_{22}	0.020	1.552
b_{33}	0.154	38.251
b_{12}	0.110	22.816
b_{13}	0.351	23.150
b_{23}	0.029	12.463
		17.226
		32.924
		26.013
		8.712
		17.208
		17.208
		17.208
		17.208
		17.208
		17.208

2.2.3.4 Predictions

Once $\boldsymbol{b}$ is determined, it is then possible to predict or estimate $\boldsymbol{y}$ and to calculate the sums of squares and other statistics as outlined in Sections 2.2.2 and 2.2.4. For the data in Table 2.6, the results are presented in Table 2.8, using the pseudo-inverse to obtain $\boldsymbol{b}$ and then predict $\hat{\boldsymbol{y}}$. Note that the *size* of the parameters does not necessarily relate to significance, in this example. It is a common misconception that the larger the coefficient, the more important it is. For example, it may appear that b_{22} is small (0.020) relative to the b_{11} parameter (0.598), but this depends on the physical measurement units.

- The pH range is between 4 and 6; hence, the square of pH varies between 16 and 36 or by 20 units overall.
- The temperature range is between 20 and 60 °C, the squared range varying between 400 and 3600 or by 3200 units overall, which is a 160-fold difference in range compared with pH.
- Therefore, to be of equal importance, b_{22} would need to be 160 times smaller than b_{11}.
- As the ratio between of b_{11}:b_{22} is 29.95, b_{22} is in fact considerably more significant than b_{11}.

In Section 2.2.4, we discuss in more detail how to tell whether a given parameter is significant, but it is indeed very dangerous to rely on visual inspection of tables of regression parameters and make deductions from these without understanding carefully how the data are scaled.

If carefully interpreted, three types of information can come from the model.

- The size of the coefficients can help the experimenter decide how significant the coefficient is, although the absolute value may be misleading in some cases. For example, does a change in pH significantly change the yield of a reaction? Or is the interaction between pH and temperature significant? In other words, does the temperature at which the reaction has a maximum yield differ substantially at pH 5 and at pH 7?
- The coefficients can be used to construct a model of the response, for example, the yield of a reaction as a function of pH and temperature, and to establish the optimum conditions for obtaining the best yield. In this case, the experimenter is not so interested in the precise equation for the yield but is very interested in the best pH and temperature.
- Finally, a quantitative model may be interesting. Predicting the concentration of a compound from the absorption in a spectrum requires an accurate knowledge of the relationship between the variables. Under such circumstances,

Table 2.9 Coding of data.

Variable	Units	−1	+1
pH	$-\log[H^+]$	4	6
Temperature	°C	20	60
Concentration	mM	2	4

the value of the coefficients is important. In some cases, it is known that there is a certain kind of model, and the task is primarily to obtain a regression or calibration equation.

Although the emphasis in this chapter is on using MLR techniques, it is important to recognise that the analysis of designed experiments is not restricted to such approaches, and it is legitimate to employ multivariate methods such as principal components regression and PLS as described in detail in Chapter 6.

1. The term b_1 in a model is estimated at 0.736 and b_2 at 5.129. Does this imply b_1 is less significant than b_2?
   (a) Yes
   (b) Cannot tell unless the design matrix is known
   (c) Cannot tell unless the replicate error is known.

2.2.4 Assessment of Significance

In many traditional books on statistics and analytical chemistry, large sections are devoted to significance testing. Indeed, an entire and very long book could easily be written about the use of significance tests in chemistry. However, much of the work on significance testing goes back nearly a 100 years, to the work of 'Student' (a pseudonym for W.S. Gossett), and slightly later to R.A. Fisher. Although their methods based primarily on the *t*-test and *F*-test have had a huge influence in applied statistics, they were developed before the modern computer age. A typical statistical calculation using pen and paper and perhaps a book of logarithm or statistical tables might take several days, compared with a few seconds on a modern microcomputer. Ingenious and elaborate approaches were developed, including special types of graph papers and named methods for calculating the significance of various effects.

These early methods were developed primarily for use by specialised statisticians, mainly trained as mathematicians, in an environment where user-friendly graphics or easy analysis of data was inconceivable. A mathematical statistician will have a good feeling for the data and so is unlikely to perform calculations or compute statistics from a data set unless satisfied that the quality of data is appropriate. In the modern age, everyone can have access to these tools without a great deal of mathematical expertise, but correspondingly, it is possible to misuse these methods in an inappropriate manner. The practicing chemist needs to have a numerical and graphical feel for the significance of his or her data, and traditional statistical tests are only one of a battery of approaches to determine the significance of a factor or effect in an experiment.

This section provides an introduction to a variety of approaches for assessing significance. For historic reasons, some methods such as cross-validation and independent testing of models are best described in the chapters on multivariate methods (see Chapters 4–6), although the chemometrician should have a broad appreciation of all such approaches and not be restricted to any one set of methods.

2.2.4.1 Coding

In Section 2.2.3, we introduced an example of a three-factor design, given in Table 2.6, described by 10 terms in the corresponding model. Our comment was that the significance of the coefficients cannot easily be assessed by inspection because the physical scale for each variable is different. In order to have a better idea of the significance, it is useful to put each variable on a comparable scale. It is common to *code* experimental data. Each variable is placed on a common scale, often with the highest coded value of each variable equal to +1 and the lowest to −1. Table 2.9 represents a possible way to scale the data; hence, for factor 1 (pH), a coded value (or level) or −1 corresponds to a true pH of 4. Note that coding does not need to be linear: in fact, pH is actually measured on a logarithmic scale; hence, we are not coding $[H^+]$ using a linear scale.

2.2.4.2 Size of Coefficients

The simplest approach to determining significance is to simply look at the magnitude of the terms (or coefficients) in the model. Provided the data is coded correctly, the larger the term, the greater its significance. This depends on each coded factor varying over approximately the same range (between +1 and −1 in this case). Clearly, small differences in range are not important, often the aim is to say whether a particular term has a significant influence or not, rather than a detailed interpretation of their size. A value of 5.343 for b_1 implies that on average, in the centre of the design, the response is higher by 5.343 if the value of b_1 is increased by one coded pH unit. This is easy to verify and provides an alternative, classical approach to the calculation of the coefficients.

- Consider the 10 experiments in Table 2.6 at which b_1 is at a coded level of either +1 or −1, namely the first 10 experiments.
- Then, group these in five pairs, each of which the levels of the other two main factors are identical. These pairs are {1,5}, {2,6}, {3,7}, {4,8} and {9,10}.
- Take the difference between the responses at the levels and average them

$$[(34.841 - 19.825) + (16.567 - 1.444) + (45.396 - 37.673) + (27.939 - 23.131) + (23.088 - 12.325)]/5,$$

  which gives an answer of 10.687, representing the average change in value of the response when the pH is increased from a coded value of −1 to one of +1, half of which equals the coefficient 5.343.
- Note that this calculation is only valid because b_1 is at its central value (i.e. a coded value of 0) for the remaining 10 experiments. It will not be valid for the b_2 and b_3 unless different experiments are chosen.

It is useful to make practical deductions from the data, which will guide the experimenter.

- The response varies over a range of 43.953 units between the lowest and highest observation in the experimental range.
- Hence, the linear effect of pH, on average, is to increase the response by twice the coded coefficient or 10.687 units over this range, approximately 25% of the variation, probably quite significant. The effect of the interaction between pH and concentration (b_{13}), however, is only 0.702 units or a relatively small contribution, rather less than the replicate error; hence, this factor is unlikely to be significant.
- The squared terms must be interpreted slightly differently. The lowest possible coded value for the squared terms is 0, not −1; hence, we do not double these values to obtain an indication of significance, the range of variation of the squared terms being between 0 and +1, or half that of the other terms.

It is not, of course, necessary to have replicates to perform this type of analysis. If the yield of a reaction varies between 50% and 90% over a range of experimental conditions, then a factor that contributes, on average, to only 1% of this increase is unlikely to be too important. However, it is vital that the factors are coded for meaningful comparison. In addition, certain important properties of the design (namely orthogonality), which will be discussed in detail in later sections, are also important.

Provided the factors are coded correctly, it is quite easy to make qualitative comparisons of significance simply by examining the size of the coefficients either numerically and graphically. In some cases, the range of variation of each individual factor might differ slightly (e.g. squared and linear terms above), but provided this is not dramatic, for rough indications, the size of the factors can be legitimately compared. In the case of two-level factorial designs (described in Sections 2.3.1–2.3.3), each factor is normally scaled between −1 and +1; hence, all coefficients are on the same scale.

1. A design is coded so that a temperature of 15 °C is −1 and 55 °C +1. Temperature is donated as factor 1 and $b_1 = +8\%$ of the range of yields of a reaction over the design. This implies, on average, in the centre of the design
   (a) We expect the yield to increase at the rate of 8% every 10 °C
   (b) We expect the yield to increase at the rate of 4% every 10 °C
   (c) We expect the yield to increase at the rate of 16% every 10 °C.

2.2.4.3 Student's *t*-Test

An alternative, statistical indicator based on Student's *t*-test can be used, provided more experiments are performed than the terms in the model. Although this and related statistical indicators have a long and venerated history, it is always important to back up the statistics by simple graphs and considerations about the data. There are many diverse applications of a *t*-test, but in the context of analysing the significance of factors on designed experiments, the following main steps are used and are illustrated in Table 2.11 for the example described above using the coded values in Table 2.9.

the significance, the smaller the percentage, so simply subtract from 100). The higher this probability, the greater the confidence that the factor or term is significant. Hence, using Table A.3, we see that a critical value of 4.1437 indicates 99.9% certainty and that a parameter is significant for 10 degrees of freedom; hence, any value above this is highly significant. A 95% significance results in a value of 1.8125, so b_{23} is just above this level. In fact, the numbers in Table 2.11 were calculated using the Excel function TDIST, which provides probabilities for any value of t and any number of degrees of freedom. Normally, quite high probabilities are expected if a factor is significant, often in excess of 95%.

- In some articles, p values are used instead of percentage confidence. Analytical chemists tend to like the idea of confidence but statisticians like the idea of p values, for historic reasons. A p value of 0.05 corresponds to a percentage confidence of 95%. Strictly speaking, we talk about the probability of rejecting the null hypothesis. The null hypothesis test whether any given term has no significance in the model. If a p value is calculated to be 0.01, this means that there is a 1% chance of rejecting the null hypothesis, or that it is very unlikely the term has no significance, but chemists often like to say that there is 99% confidence that the term is significant. Additionally, this confidence or p value assumes that the underlying sample errors are normally distributed, an assumption that is often only roughly obeyed in practice; however, low p values or high confidence will usually suggest that a term or factor should be retained.

1. The p value for a term is calculated to be 0.02. This implies that it is not significant.
   (a) True
   (b) False
2. A design consists of 15 experiments in total, and a 10-term model is formed. How many degrees of freedom are remaining for determining the confidence in the regression model?
   (a) 10
   (b) 5
   (c) 1
3. The t-test can be employed to determine the significance of terms in a full factorial model for a full factorial design.
   (a) True
   (b) False

2.2.4.4 *F*-Test

The F-test is another alternative. A common use of the F-test is together with ANOVA and determines the significance of one variance (or mean sum of squares) in relation to another. Typically, the lack-of-fit is compared with the replicate error. Simply determine the ratio S_{lof}:S_{rep} (e.g. see Table 2.4) and check the size of this number. F-distribution tables are commonly presented at various probability levels, which are often called *confidence levels*. We use a one-tailed F-test in this case as the aim is to see whether one variance is significantly bigger than another, not whether it differs significantly; this differs from the t-test that is two tailed in the application described in Section 2.2.4.3. The columns correspond to the number of degrees of freedom for S_{lof} and the rows to S_{rep} (in the case discussed in here). The table allows to determine the significance of the error (or variance) represented along the columns relative to that represented along the rows. Consider the proposed models for data sets A and B, both excluding the intercept. Locate the relevant number (for a 95% confidence, the lack-of-fit is significant – or for a p value of 0.05, the null hypothesis is rejected, with five degrees of freedom for the lack-of-fit and four degrees of freedom for the replicate error, this number is 6.26, see Table A.5 (given by a distribution often called $F_{(5,4)}$ see Section A.3.5) – hence, an F-ratio must be greater than this value for this level of confidence). Returning to Table 2.4, it is possible to show that the chances of lack-of-fit to a model without an intercept are not very high for the data in Figure 2.9 (ratio = 0.49); however, there is some doubt about the data arising from Figure 2.10 (ratio = 1.79); using the FDIST function in Excel, we can see that the probability is 70.4%, below the 95% confidence the intercept is significant (or in other words with a $p = 0.396$, the null hypothesis is correct, so over one time in three if there were no underlying significance in the intercept, we will still expect to obtain this value of F or more), but still high enough to give us some doubts. Nevertheless, the evidence is not entirely conclusive because the intercept term (2.032) is approximately the same order of magnitude as the replicate error (1.194); for this level of experimental variability, it will never be possible to predict and model the presence of an intercept of this size with a high degree of confidence.

Table 2.12 *F*-ratio for experiment with low experimental error.

Concentration	Absorbance		Model with intercept	Model without intercept
1	3.500	b_0	0.854	n/a
1	3.398	b_1	2.611	2.807
2	6.055			
3	8.691	S_{reg}	0.0307	1.4847
3	8.721	S_{rep}	0.0201	0.0201
4	11.249	S_{lof}	0.0107	1.4646
4	11.389			
5	13.978			
6	16.431			
6	16.527	*F*-ratio	0.531	58.409

The solution is to perform new experiments, perhaps on a different instrument, in which the reproducibility is much greater. Table 2.12 is of such a data set, with essential statistics indicated. Now the *F*-ratio for the lack-of-fit without the intercept becomes 58.409, which is significant at the >99% level (critical value from Table A.4), whereas the lack-of-fit with the intercept included is less than the experimental error.

1. $F_{(6,3)}$ is equal to $F_{(3,6)}$.
   (a) True
   (b) False
2. It is essential to have replicate measurements for both an *F*-test and a *t*-test to determine the significance of terms in a model using the methods described in this chapter.
   (a) Yes.
   (b) No, only for an *F*-test.
   (c) No, only for a *t*-test.
   (d) No, for neither.

2.2.4.5 Normal Probability Plots

For designs where there are no replicates (essential for most uses of the *F*-test) and also where there no degrees of freedom available to assess the lack-of-fit to the data (essential for a *t*-test), other approaches can be employed to examine the significance of coefficients.

As will be discussed in Section 2.3, two-level factorial designs are common, and providing the data are appropriately coded, the size of the coefficients directly relates to their significance. Normally, several terms are calculated, and an aim of modelling is to determine which have significance, the next step, possibly being to then perform another, a more detailed design for quantitative modelling of the significant effects. Often it is convenient to present the coefficients graphically, and a classical approach is to plot them on a normal probability paper. Before the computer age, a large number of different types of statistical graph paper would be available, assisting data interpretation. However, nowadays, it is quite easy to obtain relevant graphs using simple computer packages.

The principle of normal probability plots is that in the absence of any systematic trend, a randomly generated set of numerical values will fall into a normal distribution (see Section A.3.2). Hence, if, for example, we look at the size of seven effects, for example, as assessed by their values of *b* (provided the data are properly coded and the experiment is well designed, of course), and the effects are simply random, on average, we would expect the probable size of each effect to be normally distributed. In Figure 2.14, seven lines are indicated on the normal distribution curve (the horizontal axis representing the number of standard deviations from the mean) so that the areas between each line equal 1/7 the total area (the areas at the extremes adding up to 1/7 in total). If, however, an effect is very large, it will fall at a very high or low value, so large that it is unlikely to be the consequence of a random process, and is significant. Normal probability plots can be used to rank the coefficients in size (the most negative being the lowest, the most positive being the highest), from the rank determining the expected position in the normal probability plot and then

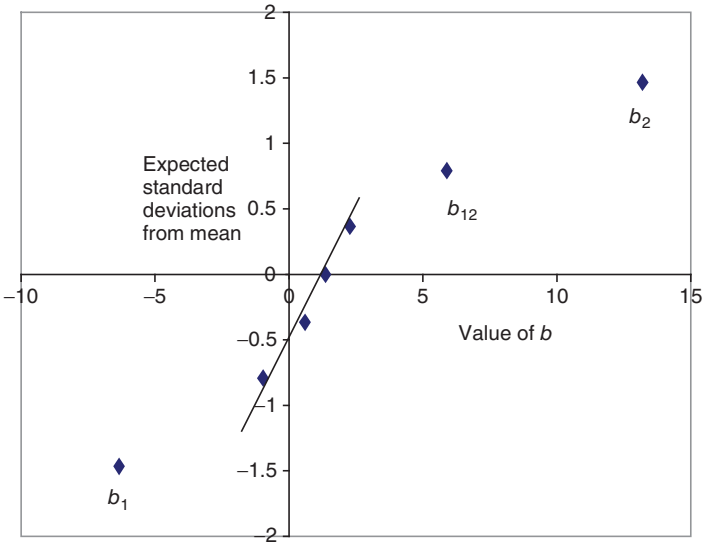


Figure 2.15 Normal probability plot for data in Table 2.13 with significant factors marked.

in certain cases, it can provide useful preliminary graphical information, although probably is not much used in modern computer-based chemometrics.

1. Normal probability plots can be used where there are no replicates and no degrees of freedom to determine lack of fit in a model.
   (a) True
   (b) False

2.2.4.6 Dummy Factors

Another very simple approach is to include one or more dummy factors. These can be built into a design and might, for example, be the colour of shoes worn by the experimenter, some factor that is not likely to have a real effect on the experiment: level −1 might correspond to black shoes and level +1 to brown shoes. Mathematical models can be built including this factor, and effects smaller than this are ignored (remembering as ever to ensure that the coding of the data is sensible).

1. The size of the term arising from the dummy factor is same size as the replicate error.
   (a) True
   (b) False

2.2.4.7 Limitations of Statistical Tests

Although many traditionalists often enjoy the security statistical significance tests give, it is important to recognise that these tests do depend on assumptions about the underlying data that may not be correct, and a chemist should beware of making decisions based only on a probability obtained from a computerised statistical software package, without looking at the data, often graphically. Some typical drawbacks are as follows.

- Most statistical tests assume that the underlying samples and experimental errors are normally distributed. In some cases, this is not so; for example, when analysing some analytical signals, it is unlikely that the noise distribution will be normal; it is often determined by electronics and sometimes even data pre-processing such as the common logarithmic transform used in electronic absorption and infrared spectroscopy. In such circumstances, it may be possible to transform the data first before performing statistical analysis.

- The tests assume that the measurements arise from the same underlying population. Sometimes this is not the case, and systematic factors will come into play. A typical example involves calibration curves. It is well known that the performance of an instrument can vary from day to day. Hence, an absorption coefficient measured on Monday morning is not necessarily the same as the measured coefficient on Tuesday morning; yet, all the coefficients calculated on Monday morning might fall into the same class. If a calibration experiment is performed over several days or even hours, the performance of the instrument may vary and one solution may be to make a very large number of measurements over a long timescale, which may be impractical. Another trick is to randomise the sequence of experiments, but then it may be important to test additional effects such as the day the samples were recorded, just to work out whether this has a significant influence on the experimental data.
- The precision of an instrument must be considered. Many typical measurements, for example, in atomic spectroscopy, are recorded to only two significant figures. Consider a data set in which about 95% of the readings were recorded between 0.10 and 0.30 absorbance units; yet, a statistically designed experiment is used to estimate 64 effects. The *t*-test provides information on the significance of each effect. However, statistical tests assume that the data are recorded to a high degree of accuracy and will not take this lack of numerical precision into account. For the obvious effects, chemometrics will not be necessary, but for less obvious effects, the statistical conclusions may be invalidated because of the low numerical accuracy in the raw data.

Often it is sufficient simply to look at the size of factors, the significance of the lack-of-fit statistics, perform quite simple ANOVA or produce a few graphs to make quite valid scientific deductions. In most cases, significance testing is used primarily for a preliminary modelling of the data and detailed modelling should be performed after eliminating those factors that are deemed unimportant. It is not necessary to have a very detailed theoretical understanding of statistical significance tests before the design and analysis of chemical experiments, although a conceptual appreciation of, for example, the importance of coding is essential.

1. For a series of measurements, the time they were performed can be included as a factor in the experimental design.
   (a) True
   (b) False

2.2.5 Leverage and Confidence in Models

An important experimental question relates to how well quantitative information can be predicted after a series of experiments has been carried out. For example, if observations have been made between 40 and 80 °C, what can we predict about the experiment at 90 °C? It is traditional to sharply cut off the model outside the experimental region, so that the model is used to predict only within the experimental limits. However, this approach misses much information. The ability to make a prediction often reduces smoothly from the centre of the experiments, being best at 60 °C and worse the farther away from the centre as described in the example above. This does not imply that it is impossible to make any statement about the response at 90 °C, simply that there is less confidence in the prediction than at 80 °C, which, in turn, is predicted less well than at 60 °C. It is important to be able to visualise how the ability to predict a response (e.g. a synthetic yield or a concentration) varies as the independent factors (e.g. pH and temperature) are changed.

When only one factor is involved in the experiment, the predictive ability is often visualised by confidence bands. The 'size' of these confidence bands depends on the magnitude of the experimental error. The 'shape', however, depends on the experimental design and can be obtained from the design matrix (Section 2.2.3) and is influenced by the arrangement of experiments, replication procedure and mathematical model. The concept of *leverage* is used as a measure of such confidence. The mathematical definition is given by

$$\boldsymbol{H} = \boldsymbol{D}(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{D}'$$

where $\boldsymbol{D}$ is the design matrix. This new matrix is sometimes called the *hat matrix* and is a symmetric square matrix with the number of rows and columns equal to the number of experiments. Each experimental point n has a corresponding value of leverage h_{nn} (the diagonal element of the hat matrix) associated with it. Alternatively, the value of leverage for experiment n can be calculated as follows:

$$\boldsymbol{h}_{nn} = \boldsymbol{d}_n(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{d}_n'$$

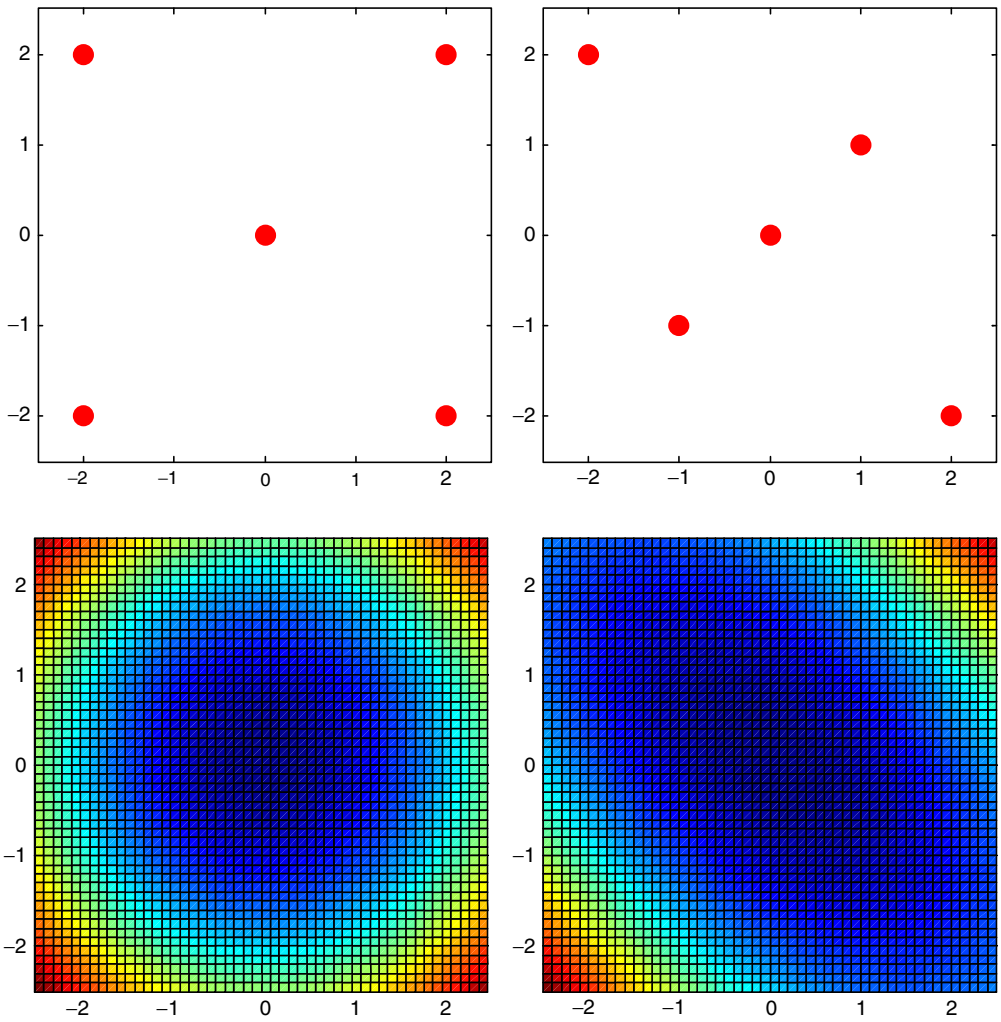


Figure 2.19 Two experimental arrangements together with the corresponding leverage for a linear model.

2. The hat matrix is always a diagonal matrix
   (a) True
   (b) False

3. If we know how far an experiment is from the centre of a design and the mathematical model used, we can then calculate leverage.
   (a) True
   (b) False

2.3 Factorial Designs

In this and the remaining sections of this chapter, we will introduce a number of possible designs, which can be understood using the building blocks introduced in Section 2.2. Factorial designs are some of the simplest, often used for screening or when there are a large number of possible factors. As will be seen, they have limitations but are the easiest to understand. Many designs are presented as a set of rules that provide the experimenter with a list of conditions, below we will present the rules for many common methods.

2.3.1 Full Factorial Designs

Full factorial designs at two levels are mainly used for screening, which is to determine the influence of a number of effects on a response and to eliminate those that are not significant, the next stage being to undertake a more detailed study. Sometimes where detailed predictions are not required, the information from factorial designs is adequate, at least in situations where the aim is fairly qualitative (e.g. to improve the yield of a reaction rather than obtain a highly accurate rate dependence that is then interpreted in fundamental molecular terms).

Consider a chemical reaction whose performance is known to depend on pH and temperature, including their interaction. A set of experiments can be proposed to study these two factors, each at two levels (e.g. two temperatures), using a two-level, two-factor experimental design. The number of experiments is given by $N = l^k$, where l is the number of levels (=2) and k the number of factors (=2); hence, in this case, N equals 4. For three factors, the number of experiments will equal 8 and so on, provided the design is performed only at two levels. The following stages are used to construct the design and interpret the results.

- The first step is to choose a high and low level for each factor; for example, if temperature is factor 1, use 30 and 60 °C, and if pH is factor 2, use 4 and 6. This choice will require the experimenter's judgment to determine sensible ranges.
- The next step is to use a standard design. The value of each factor is usually coded (see Section 2.2.4.1) as '−' (low) or '+' (high). Note that some authors use −1 and +1 or even 1 and 2 for low and high. When reading different texts, do not get confused; always first understand what notation has been employed. There is no universally agreed convention for coding; however, design matrices that are symmetric around 0 are almost always easier to handle computationally. There are four possible unique sets of experimental conditions that can be represented as a table analogous to four binary numbers 00 (−−), 01 (−+), 10 (+−) and 11 (++), which can be related to a set of physical conditions. For example, the set of experiments represented by −+ involve performing an experiment at 30 °C and pH 6 in our case.
- Next, perform the experiments and obtain the response under each set of experimental conditions. Table 2.16 illustrates the coded and true set of experimental conditions plus the corresponding response, which might, for example, be the percentage of a by-product, the lower the better. Something immediately appears strange from these results. Although it is obvious that the higher the temperature, the higher the percentage by-product, there does not at first seem to be any consistent trend as far as pH is concerned. Provided the experimental results were recorded correctly, this suggests that there must be an interaction between temperature and pH. At a lower temperature, the percentage decreases with an increase in pH, but the opposite is observed at a higher temperature. How can we interpret this?
- The next step, of course, is to analyse the data, by constructing a design matrix (Section 2.2.3). We know that an interaction term must be taken into account and can set up a design matrix as given in Table 2.17 based on a model of the

Table 2.16 Coding of a simple two factor, two level design and corresponding responses.

Experiment number	Factor 1	Factor 2	Temperature	pH	Response
1	−	−	30	4	12
2	−	+	30	6	10
3	+	−	60	4	24
4	+	+	60	6	25

Table 2.17 Design matrix.

Intercept	Temperature	pH	Temp. × pH
1	30	4	120
1	30	6	180
1	60	4	240
1	60	6	360

1	x_1	x_2	$x_1 x_2$
+	−	−	−
+	−	+	−
+	+	−	+
+	+	+	+

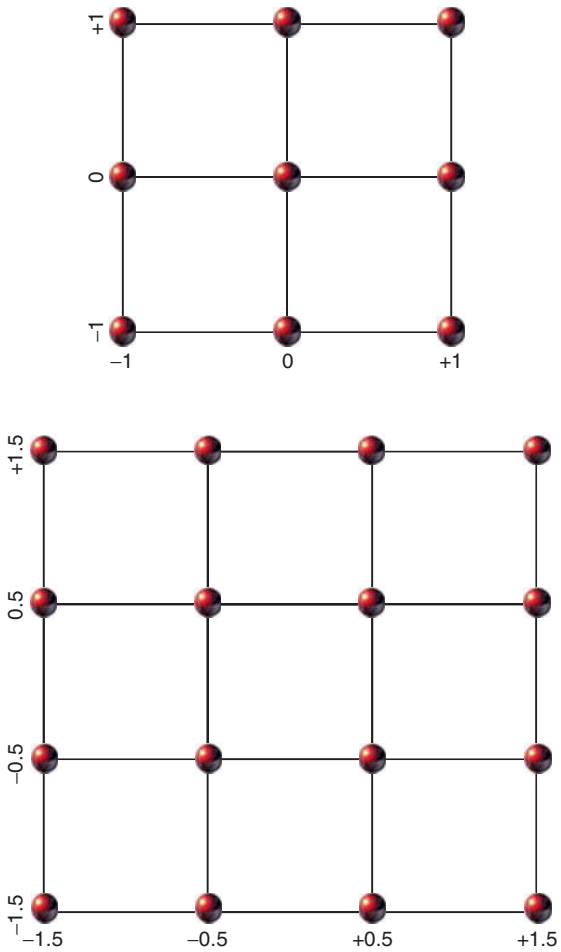


Figure 2.21 Three- and four-level full factorial designs.

3. It is desired to form a two-level, five-factor full factorial design and model the data using just the intercept, linear and two-factor interaction terms. How many degrees of freedom are available to determine fit to the model?
   (a) 32
   (b) 16
   (c) 10

2.3.2 Fractional Factorial Designs

A weakness of full factorial designs is the large number of experiments that must be performed. For example, for a 10-factor design at two levels, 1024 experiments are required, which may be impracticable. The large number of experiments does not always result in useful or interesting extra information; hence, performing all these result in waste of time and resources. Especially in the case of screening, where a large number of factors may be of potential interest, it is inefficient to run so many experiments in the first instance. There are numerous tricks to reduce the number of experiments.

Consider a three-factor, two-level design. Eight experiments are listed in Table 2.21, the conditions being coded as usual. Figure 2.22 is a symbolic representation of the experiments, often presented on the corners of a cube, whose axes correspond to each factor. The design matrix for all the possible coefficients that can be set up as is also presented in Table 2.21 and consists of eight possible columns, equal to the number of experiments. Some columns represent interactions, such as the three-factor interaction, that may not be very significant. At first screening, we may primarily wish to say whether the three factors have any real influence on the response, not to study the model in detail. In a more complex situation, we may wish to screen 10 possible factors; hence, reducing the number of factors to be studied in

2.3.3 Plackett–Burman and Taguchi Designs

Where the number of factors is quite large, the constraint that the number of experiments must be equal to a power of 2 can be rather restrictive. As the number of experiments must always exceed the number of factors by at least one (to take into account the intercept), this would mean, for example, that 32 experiments are required for the study of 19 factors and 64 experiments for the study of 43 factors. In order to overcome this problem and reduce the number of experiments, other approaches are needed.

Plackett and Burman published their classic paper in 1946, which has been much cited by chemists. Their work originated from the need for wartime testing of components in equipment manufacture. A large number of factors influenced the quality of these components; thus, efficient procedures were required for screening. They proposed a number of two-level factorial designs, where the number of experiments is a multiple of 4. Hence, designs exist for $N = 4$, 8, 12, 16, 20, 24 and so on experiments. The number of experiments exceeds the number of factors, k, by 1, hence available for factors 3, 7, 11 and so on.

One such design is given in Table 2.26 for 11 factors and 12 experiments and has various features.

- In the first row, all factors are at the same level, in this case at the low (−) level.
- The first column from rows 2 to k is called a *generator*. The key to the design is that there are only certain allowed generators that can be obtained from tables. Note that the number of factors will always be an odd number equal to $k = 4m - 1$ (or 11 in this case), where m is any integer. If the first row consists of '−', the generator will consist of $2m$ (=6 in this case) experiments at '+' level and $2m - 1$ (=5 in this case) at '−' level, the reverse being true if the first row is at the '+' level. In Table 2.26, the generator is + + − + + + − − − + −, as outlined by a box.
- The next $4m - 2$ (=10) columns are generated from the first column simply by shifting the down cells by one row. This is indicated by diagonal arrows in the table. Notice that experiment 1 is not included in this procedure.
- The level of factor j in experiment (or row) 2 equals to the level of this factor in the row k for factor $j - 1$. For example, the level of factor 2 in experiment 2 equals the level of factor 1 in experiment 12.

There are as many high as low levels of each factor over the 12 experiments, as would be expected. The most important property of the design, however, is *orthogonality*. Consider the relationship between factors 1 and 2.

- There are six instances in which factor 1 is at a high level, and factor 6 at a low level.
- For each of the six instances at which factor 1 is at a high level, factor 2 is at a high level in three cases and it is at a low level in the other three cases. A similar relationship exists where factor 1 is at a low level. This implies that the factors are orthogonal or uncorrelated, an important condition for a good design.
- Any combination of two factors is related in a similar way.

Only certain generators possess all these properties; hence, it is important to use only known generators.

Standard Plackett–Burman designs exist for 7, 11, 15, 19 and 23 factors; generators are given in Table 2.27. A few of the designs for more than 23 factors are slightly more complicated to construct, for which readers should refer to their paper of 1946. Note that for 7 and 15 factors, it is also possible to use fractional factorial designs as discussed in Section 2.3.2. However, in the old adage, 'all roads lead to Rome'; in fact, fractional factorial and Plackett–Burman designs are equivalent, the difference simply being in the way the experiments and factors are organised in the data

Table 2.26 A Plackett–Burman design for 11 factors, generator outlined by a box.

Experiments \ Factors	1	2	3	4	5	6	7	8	9	10	11
1	−	−	−	−	−	−	−	−	−	−	−
2	+	−	+	−	−	−	+	+	+	−	+
3	+	+	−	+	−	−	−	+	+	+	−
4	−	+	+	−	+	−	−	−	+	+	+
5	+	−	+	+	−	+	−	−	−	+	+
6	+	+	−	+	+	−	+	−	−	−	+
7	+	+	+	−	+	+	−	+	−	−	−
8	−	+	+	+	−	+	+	−	+	−	−
9	−	−	+	+	+	−	+	+	−	+	−
10	−	−	−	+	+	+	−	+	+	−	+
11	+	−	−	−	+	+	+	−	+	+	−
12	−	+	−	−	−	+	+	+	−	+	+

Western practices. His philosophy was that consumers desire products that have constant properties within narrow limits. For example, a consumer panel may taste the sweetness of a product, rating it from 1 to 10. A good marketable product may result in a taste panel score of 8: above this, the product is too sickly, and below, the consumer expects the product to be sweeter. There will be a huge number of factors in the manufacturing process that might cause deviation from the norm, including suppliers of raw materials, storage and preservation of the food and so on. Which factors are significant? Taguchi developed designs for screening large number of potential factors.

His designs are presented in the form of table similar to that of Plackett and Burman, but with a '1' for a low and '2' for a high level. Superficially, Taguchi's designs might appear different, but by changing the notation, and swapping rows and columns around, it is possible to show that both types of design are identical, and, indeed, the simpler designs are the same as the well-known fractional factorial designs. There is a great deal of controversy surrounding Taguchi's work; although many statisticians feel that he has reinvented the wheel, he was an engineer, and his way of thinking had a major and positive effect on Japanese industrial productivity. Before globalisation and the Internet, there was less exchange of ideas between different cultures. His designs are part of a more comprehensive approach to quality control in industry.

Taguchi's designs can be extended to three or more levels, but construction becomes fairly complicated. Some texts do provide tables of multi-level screening designs, and it is also possible to mix the number of levels, for example, having one factor at two levels and another at three levels. This could be useful, for example, if there are three alternative sources of one raw material and two of another raw material. Remember that the factors can fall into discrete categories and do not have to be numerical values such as temperature or concentrations. A large number of designs have been developed from Taguchi's work, but most are quite specialist, and it is not easy to generalise. The interested reader is advised to consult the source literature.

1. For seven factors, a quarter factorial and a Plackett–Burman design are the same apart from rearrangement of rows and/or columns.
   (a) True
   (b) False
2. How many experiments are necessary for the study of 10 factors, using a Plackett–Burman design?
   (a) 12
   (b) 11
   (c) 10
3. If we want to study the effects of 13 factors using a Plackett–Burman design, how many dummy factors are necessary?
   (a) 0
   (b) 1
   (c) 2

2.3.4 Partial Factorials at Several Levels: Calibration Designs

Two-level designs are useful for exploratory purposes and can sometimes result in quite useful models, but in many areas of chemistry, such as calibration (see Chapter 5 for more details), it is desirable to have several levels, especially when studying mixtures. Much of chemometrics is concerned primarily with linearly additive models of the expected form $\boldsymbol{X} = \boldsymbol{C}\,\boldsymbol{S}$, where $\boldsymbol{X}$ is an observed matrix, such as a set of spectra, each row consisting of a spectrum and each column of a wavelength, $\boldsymbol{C}$ is a matrix of, for example, concentrations, each row consisting of the concentration of a number of compounds in a spectrum, and $\boldsymbol{S}$ could consist of the corresponding spectra of each compound. There are innumerable variations on this theme; in some cases, all the concentrations of all the components in a mixture are known, whereas in other cases, the concentrations of only a few components in a mixture are known. In many situations, it is possible to control the experiments by mixing up components in the laboratory, but in other cases, this is not practicable, samples are being taken from the field. A typical laboratory-based experiment might involve recording a series of four component mixtures in which each component is studied at five concentration levels.

A recommended strategy is as follows:

- Perform a calibration experiment, by producing a set of mixtures of a series of compounds of known concentrations to give a 'training set'.

Table 2.30 Parameters for construction of a multi-level calibration design.

Levels	Experiments	Max number Orthogonal factors	Repeater	Difference vectors	Cyclic permuters
3	9	4	Any	{01}, {10}	
4	16	5	Any	{021}, {120}	
5	25	12	0	{0231}, {1320}, {2013}, {3102}	$-2 \rightarrow -1 \rightarrow 2 \rightarrow 1 \rightarrow -2$, $-2 \rightarrow 1 \rightarrow 2 \rightarrow -1 \rightarrow -2$
7	49	16	0	{241035}, {514302}, {451023}, {124350}, {530142}, {203415}, {320154}, {053421}	$-3 \rightarrow 2 \rightarrow 3 \rightarrow -1 \rightarrow 1 \rightarrow -2 \rightarrow -3$, $-3 \rightarrow 1 \rightarrow -1 \rightarrow 2 \rightarrow 3 \rightarrow -2 \rightarrow -3$, $-3 \rightarrow -2 \rightarrow 3 \rightarrow 2 \rightarrow -1 \rightarrow 1 \rightarrow -3$, $-3 \rightarrow -2 \rightarrow 1 \rightarrow -1 \rightarrow 3 \rightarrow 2 \rightarrow -3$

to expand the number of compounds in the mixture, involving twice the number of experiments for a doubling of the number of factors.

1. It is possible to set up a calibration design for three levels involving 18 experiments.
   (a) True
   (b) False
2. For each factor, only one experiment is performed at the repeater level for all factors in a calibration design.
   (a) True
   (b) False
3. It is possible to set up a calibration design using six levels
   (a) True
   (b) False

2.4 Central Composite or Response Surface Designs

Two-level factorial designs are primarily useful for exploratory purposes, and calibration designs have special uses in areas such as multivariate calibration, where we often expect an independent linear response from each component in a mixture. It is often, though, important to provide a more detailed model of a system. There are two prime reasons. The first is for optimisation – to find the conditions that result in a maximum or minimum as appropriate. An example is when improving the yield of synthetic reaction or a chromatographic resolution. The second is to produce a detailed quantitative model: to predict mathematically how a response relates to the values of various factors. An example may be to know how the near-infrared spectrum of a manufactured product relates to the nature of the material and process employed in manufacturing.

Most exploratory designs do not involve recording replicates, nor do they provide information on squared terms; some such as Plackett–Burman and highly fractional factorials do not even provide details of interactions. When we want to form a detailed model, it is often desirable at a first stage to reduce the number of factors via exploratory designs, described in Section 2.3, to a small number of main factors (perhaps 3 or 4) that are to be studied in detail, for which both squared and interaction terms in the model are of interest.

2.4.1 Setting up the Design

Many designs for use in chemistry for modelling are based on the central composite design (sometimes called a *response surface design*), the main principles of which will be illustrated via a three-factor example (Figure 2.28 and Table 2.31).

- Performing a full factorial design, a star design and five replicates, results in 20 experiments. This design is a type of central composite design. When the axial or star points are situated at $a = \pm 1$, the design is sometimes also called a *face-centred cube design*, see Table 2.31. Note that the number of replicates is one less than the number of experiments performed in the centre.

1. How many experimental points are required for a four-factor star design?
   (a) 7
   (b) 4
   (c) 8
   (d) 9
2. How many experiments are required for a five-factor central composite design with five replicates in the centre?
   (a) 47
   (b) 48
   (c) 31
   (d) 32

2.4.2 Degrees of Freedom

In this section, we analyse the features of such designs in detail. However, many factors are used in most cases, only two-factor interactions are computed; hence, higher order interactions are ignored, although, of course, these provide sufficient degrees of freedom to estimate the lack-of-fit.

- The first step is to set up a model. A full model including all two-factor interactions consists of $1 + 2k + [k(k-1)]/2 = 1 + 6 + 3$ or 10 coefficients in the case of a three-factor design relating to
  – 1 intercept term (of the form b_0)
  – 3 (=k) linear terms (of the form b_1)
  – 3 (=k) squared terms (of the form b_{11})
  – and 3 (=$[k(k-1)]/2$) interaction terms (of the form b_{12})
  or in equation form

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_{11}{x_1}^2 + b_{22}{x_2}^2 + b_{33}{x_3}^2 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3$$

- A degree-of-freedom tree can be drawn up as illustrated in Figure 2.29.
  We can see that
  – there are 20 (=N) experiments overall,
  – 10 (=P) parameters in the model,
  – 5 (=R) degrees of freedom to determine replication error and
  – 5 (=$N - P - R$) degrees of freedom for the lack-of-fit.
  Note that the number of degrees of freedom for the lack-of-fit equals that for replication in this case, suggesting quite a good design.

The total number of experiments N (=20) equals the sum of

- 2^k (=8) factorial points, often represented as the corners of the cube,
- $2k + 1$ (=7) star points, often represented as axial points on (or above) the faces of the cube plus one in the centre and
- R (=5) replicate points, in the centre.

A large number of variations are present on this theme, but each design can be defined by four parameters, namely,

- The number of factorial or cubic points (N_f),
- the number of axial points (N_a), usually one less than the number of points in the star design,
- the number of central points (N_c), usually one more than the number of replicates and
- the position of the axial points a.

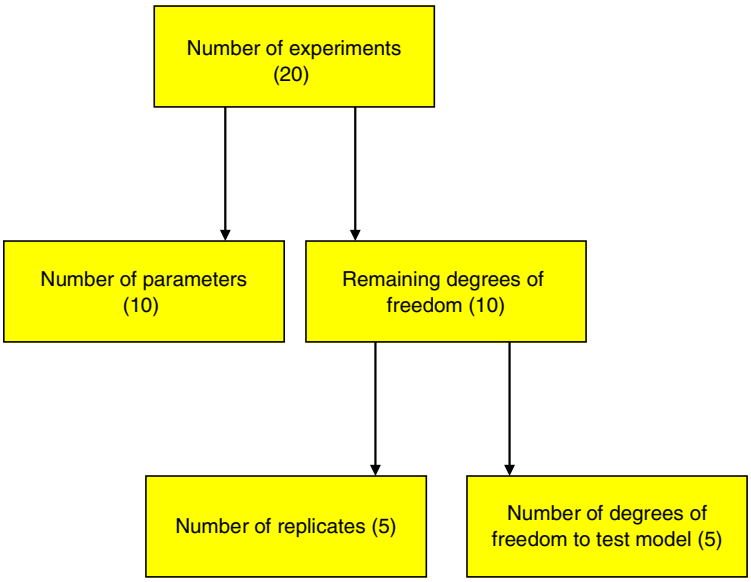


Figure 2.29 Degrees of freedom for central composite design.

In most cases, it is best to use a full factorial design for the factorial points; however, if the number of factors is large, it is legitimate to reduce this and use a partial factorial design. There are always $2k$ axial points.

The number of central points is often chosen according to the number of degrees of freedom required to assess errors via ANOVA and the F-test (see Sections 2.2.2 and 2.2.4.4) and should be approximately equal to the number of degrees of freedom for the lack-of-fit, with a minimum of about 4 unless there are special reasons for reducing this.

1. For a central composite design for four factors, using a model with the intercept, all possible linear, quadratic and two-factor interaction terms, how many degrees of freedom are there for determining the lack-of-fit error?
   (a) 6
   (b) 10
   (c) 15

2.4.3 Axial Points

The choice of the position of the axial (or star) points and how this relates to the number of replicates in the centre is an interesting consideration. Although many chemists use these designs fairly empirically, it is worth noting two statistical properties that influence the property of these designs. It is essential to recognise, although, that there is no single perfect design, indeed many of the desirable properties of a design are incompatible with each other.

- *Rotatability* implies that the confidence in the predictions depends only on the distance from the centre of the design. For a two-factor design, this means that all experimental points in a circle of a given radius will be predicted equally well. This has useful practical consequences; for example, if the two factors correspond to concentrations of acetone and methanol, we know that the farther the concentrations are from the central point (in coded values), the lower the confidence. Methods for visualising this were described in Section 2.2.5. If a design is rotatable, the confidence does not depend on the number of replicates in the centre, but only on the value of a, which should equal $\sqrt[4]{N_f}$, where N_f is the number of factorial points, equal to 2^k if a full factorial is used, for this property. Note that the position of the axial points will differ if a fractional factorial design is used.
- *Orthogonality* implies that all the terms (linear, squared and two factor interactions) are orthogonal to each other in the design matrix; that is, the correlation coefficient between any two terms (apart from the zero order term where it is not defined) equals 0. For linear and interaction terms, this will always be so, but squared terms are not so simple, and in the majority of central composite designs, they are not orthogonal. The rather complicated condition is

$a = \sqrt{\left(\sqrt{N \times N_f} - N_f\right)/2}$, which depends on the number of replicates, as a term for the overall number of experiments is included in the equation. A small lack of orthogonality in the squared terms can sometimes be tolerated, but it is often worth checking any particular design for this property.

Interestingly, these two conditions are usually not compatible, resulting in considerable dilemmas. Although in practical situations, the differences of a for the two different properties are not so large, and in some cases, it is not experimentally very meaningful to get too concerned about small differences in the axial points of the design. Table 2.32 analyses the properties of three, two-factor designs with a model of the form $y = b_0 + b_1x_1 + b_2x_2 + b_{11}x_1^2 + b_{22}x_2^2 + b_{12}x_1x_2$ ($P = 6$). Design A is rotatable, Design B is orthogonal and Design C has both properties. However, the third is inefficient in which seven replicates are required in the centre; indeed, half the design points are in the centre, which makes little practical sense, although this design is both rotatable and orthogonal. Table 2.33 lists the values of a for rotatability and orthogonality for different numbers of factors and replicates. For the five-factor design, a half factorial design is also tabulated; in all other cases, the factorial part is full. It is interesting to note that for a two-factor design with one central point (i.e. no replication), the value of a for orthogonality is 1, making it identical to a two-factor, three-level design (see Table 2.20(a)), being four factorial and five star points or 3^2 experiments in total.

Terminology varies according to authors, some calling only the rotatable designs as *true central composite designs*. It is very important to recognise that the use of statistics is very widespread throughout science, especially in experimental areas such as biology, medicine and chemistry, and check carefully an author's precise terminology. It is important not to get locked in a single textbook (even this one!), a single software package or course provider. In many cases, to simplify, a single terminology is employed from any one source. As there are no universally accepted conventions, in which chemometrics differs from, for example, organic chemistry, and most historic attempts to set up committees have come to grief or been dominated by one specific strand of opinion, every major group has its own philosophy.

The true experimental conditions can be easily calculated from a coded design. For example, if coded levels +1, 0 and −1 for a rotatable design correspond to temperatures of 30°, 40° and 50° for a two-factor design, the axial points correspond to temperatures of 25.9° and 54.1°, whereas for a four-factor design, these points are 20° and 60°. Note that these designs are only practicable where factors can be numerically defined and cannot normally be employed if some data are categorical, unlike factorial designs. However, it is sometimes possible to set the axial points at values such as ±1 or ±2 under some circumstance to allow for factors that can take discrete values, for example, the number of cycles in an extraction procedure, although this does restrict the properties of the design.

A rotatable four-factor design consists of 30 experiments, namely

- 16 factorial points at all possible combinations of ±1,
- nine star points, including a central point of (0,0,0,0) and eight points of the form (±2,0,0,0) and so on and
- typically five further replicates in the centre; note that a very large number of replicates (11) would be required to satisfy orthogonality with the axial points at 2 units, and this is probably overkilled in many real experimental situations. Indeed, if resources are available for so many replicates, it might make sense to replicate different experimental points to check whether errors are even over the response surface.

1. We know the position of the axial points for a central composite design with all factorial points.
   (a) We can tell whether the design is rotatable but do not have enough information to tell whether it is orthogonal.
   (b) We can tell whether the design is orthogonal but do not have enough information to tell whether it is rotatable.
   (c) We do not have enough information to tell whether the design is orthogonal or rotatable.
2. Not all rotatable designs are orthogonal, but all orthogonal designs are rotatable.
   (a) True
   (b) False

2.4.4 Modelling

Once the design is performed, it is then possible to calculate the values of the coefficients in the model using regression and design matrices or almost any standard statistical procedure and assess the significance of each term using ANOVA, F-tests and t-tests if felt appropriate. We can then answer, for example, whether an interaction is significant, or whether to take into account quadratic terms: these may then allow us to further simplify our model or else to ask

Table 2.33 Position of the axial points for rotatability and orthogonality for central composite designs with varying number of replicates (one less than the number of central points).

Rotatability		Orthogonality		
		R		
k		3	4	5
2	1.414	1.210	1.267	1.320
3	1.682	1.428	1.486	1.541
4	2.000	1.607	1.664	1.719
5	2.378	1.764	1.820	1.873
5 (half factorial)	2.000	1.719	1.771	1.820

questions that can be given a physical interpretation. Sometimes the significance of terms is indicated in an ANOVA table using *s, the more the greater the significance of each term (or lower p value). It is important to remember that ANOVA tables assume that the underlying errors are normally distributed.

It is also important to recognise that these designs are mainly employed in order to produce a detailed model and also to look at interactions and higher order (quadratic) terms. The number of experiments becomes excessive if the number of factors is large. If more than about five significant factors are to be studied, it is best to narrow down the problem first using exploratory designs, although the possibility of using fractional factorials on the corners helps. Remember also that it is conventional (but not always essential) to ignore interaction terms above second order.

After the experiments have been performed, it is then possible to produce a detailed mathematical model of the response surface. If the purpose is optimisation, it might then be useful, for example, by using contour or 3D plots, to determine the position of the optimum. For relatively straightforward cases, partial derivatives can be employed to solve the equations, as illustrated in Problems 7 and 16; however, if there are a large number of terms, an analytical solution can be difficult and there can also be more than one optimum. It is always recommended to try to look at the system graphically, even if there are too many factors to visualise the whole of the experimental space at once. It is also important to realise that there may be other issues that influence our definition of an optimum, such as expense of raw materials, availability of key components or even time. Sometimes a design can be used to model several responses, and each one can be analysed separately, perhaps one might be the yield of a reaction, another the cost of raw materials and another the level of impurities in a produce. Chemometricians should resist the temptation to insist on a single categorical 'correct' answer.

1. The significance of terms in a model obtained using a central composite design can be evaluated using ANOVA.
   (a) True
   (b) False

2.4.5 Statistical Factors

Another important use of central composite designs is to determine a good range of compounds for testing, as in QSAR. Consider the case in Figure 2.4. Rather than the axes being physical variables such as concentrations, they can be abstract mathematical or statistical variables such as principal components (see Chapter 4). These could come from molecular property descriptors, for example, bond lengths and angles, hydrophobicity, dipole moments and so on. Consider, for example, a database of several hundred compounds. Perhaps selection is interesting for biological tests. It may be very expensive to test all compounds; hence, a sensible strategy is to reduce the number of compounds to a selection. Taking the first two PCs as the factors, a selection of nine representative compounds can be obtained using a central composite design as follows:

- Determine the scores of the principal components of the original data set.
- Scale each PC, for example, so that the highest score equals +1 and the lowest score equals −1.

- Then, choose those compounds whose scores are closest to the desired values. For example, in the case in Figure 2.4, choose a compound whose score is closest to (−1,−1) for the bottom left-hand corner and closest to (0,0) for the centre point.
- Perform experimental tests on this subset of compounds and then use some form of modelling to relate the desired activity to structural data. Note that this modelling does not have to be multi-linear modelling, as discussed in this section, but could also be PLS, as introduced in Chapter 6.

1. It is possible for factors to be multivariate combinations of raw variables.
   (a) True
   (b) False

2.5 Mixture Designs

Chemists and statisticians use the term *mixture* in different ways. To a chemist, any combination of several substances is a mixture. In more formal statistical terms, however, a mixture involves a set of factors whose total is a constant sum, this property is often called *closure* and will be discussed in completely different contexts in the area of scaling data before principal components analysis (Sections 4.6 and 7.2.4). Hence, in statistics (and chemometrics), a solvent system in HPLC or a blend of components in products, such as paints, drugs or food, is considered a mixture, as each component can be expressed as a proportion and the total adds up to 1 or 100%. The response could be a chromatographic separation, or the taste of a foodstuff or physical properties of a manufactured material. Often the aim of modelling is to find an optimum blend of components that taste best, or provide the best chromatographic separation or the material that is most durable.

Compositional mixture experiments involve some quite specialist techniques and a whole range of considerations must be made before designing and analysing such experiments. The principal consideration is that the value of each factor is constrained. Take, for example, a three-component mixture of acetone, methanol and water, which may be solvents used as the mobile phase for a chromatographic separation. If we know that there is 80% water in the mixture, there can be no more than 20% acetone or methanol in the mixture. If there is also 15% acetone, the amount of methanol is fixed at 5%. In fact, although there are three components in the mixtures, these translate into two independent factors.

2.5.1 Mixture Space

Most chemists represent their experimental conditions in mixture space, which corresponds to all possible allowed proportions of components that add up to 100%. A three-component mixture can be represented by a triangle (Figure 2.30),

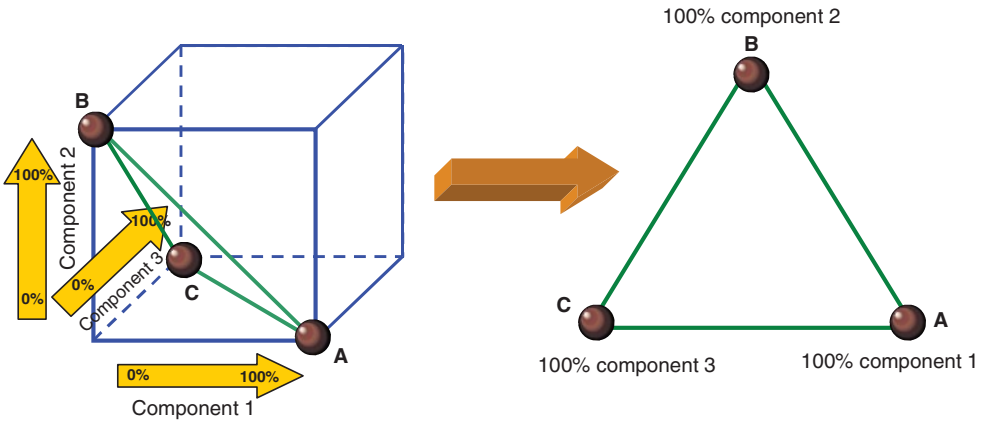


Figure 2.30 Three-component mixture space.

which is a two-dimensional cross-section of a three-dimensional space, represented by a cube, showing the allowed region in which the proportions of the three components add up to 100%. Points within this triangle or mixture space represent possible mixtures or blends.

- The three corners correspond to single components,
- points along the edges correspond to binary mixtures,
- points inside the triangle correspond to ternary mixtures,
- the centre of the triangle corresponds to an equal mixture of all the three components and
- all points within the triangle are physically allowable blends.

As the number of components increases, so does the dimensionality of the mixture space. Physically meaningful mixtures can be represented as points in this space.

- For two components, the mixture space is simply a straight line,
- for three components a triangle and
- for four components a tetrahedron.

Each object (pictured in Figure 2.31) is called a *simplex* – the simplest possible object in space of a given dimensionality: the dimensionality is one less than the number of components in a mixture, so a tetrahedron (three dimensions) represents a four-component mixture.

A number of common designs can be envisaged as ways of determining a sensible number and arrangement of points within the simplex.

1. A four-component mixture can be represented by a tetrahedron. Possible blends consisting of three of the components are represented
   (a) On the corners
   (b) On the faces
   (c) Within the tetrahedron

2.5.2 Simplex Centroid

2.5.2.1 Design

These designs are probably the most widespread. For k components, they involve performing $2^k - 1$ experiments; that is, for four components, 15 experiments are performed. It involves all possible combinations of the proportions 1, ½ to $1/k$ and is best illustrated by an example. A three-component design consists of

- three single-component combinations,
- three binary combinations and
- one ternary combination.

These experiments are represented graphically in mixture space in Figure 2.32 and tabulated in Table 2.34.

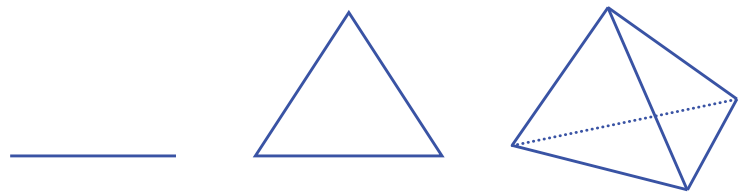


Figure 2.31 Simplex in one, two and three dimensions.

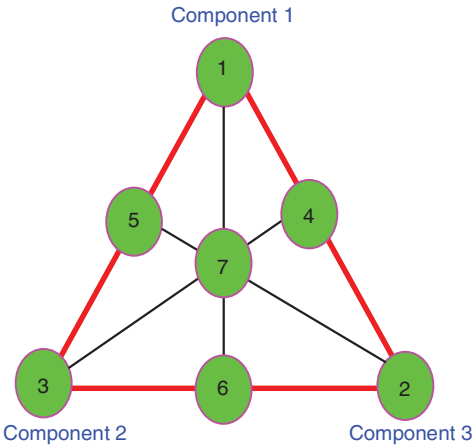


Figure 2.32 Three-component simplex centroid design.

Table 2.34 Three-component simplex centroid mixture design.

Experiment	Component 1	Component 2	Component 3	
1	1	0	0	Single component
2	0	1	0	
3	0	0	1	
4	1/2	1/2	0	Binary
5	1/2	0	1/2	
6	0	1/2	1/2	
7	1/3	1/3	1/3	Ternary

1. How many experiments are there for a four-component simplex centroid design?
   (a) 7
   (b) 12
   (c) 15

2.5.2.2 Model

Just as previously, a model and design matrix can be obtained. However, the nature of the model requires some detailed thought. Consider trying to estimate the model for a three-component design of the form

$$y = c_0 + c_1x_1 + c_2x_2 + c_3x_3 + c_{11}x_1^2 + c_{22}x_2^2 + c_{33}x_3^2 + c_{12}x_1x_2 + c_{13}x_1x_3 + c_{23}x_2x_3$$

This model consists of 10 terms, impossible if only seven experiments are performed. How can the number of terms be reduced? Arbitrarily removing three terms such as the quadratic or interaction terms has little theoretical justification. A major problem with the equation above is that the value of x_3 depends on x_1 and x_2, as it equals $1 - x_1 - x_2$ so there are, in fact, only two independent factors. If a design matrix consisting of the first four terms of the equation mentioned above was set up, it would not have an inverse, and the calculation is impossible. The solution is to set up a reduced model. Consider, instead, a model consisting only of the first three terms:

$$y = a_0 + a_1x_1 + a_2x_2$$

This is, in effect, equivalent to a model containing just the three single-component terms without an intercept as

$$y = a_0(x_1 + x_2 + x_3) + a_1x_1 + a_2x_2 = (a_0 + a_1)x_1 + (a_0 + a_2)x_2 + a_0x_3 = b_1x_1 + b_2x_2 + b_3x_3$$

It is not possible to produce a model containing both the intercept and the three single-component terms. Closed data sets, as in mixtures, have a whole series of interesting mathematical properties, but it is primarily important simply to watch for these anomalies.

The two common types of models, one with an intercept and one without, are related. Models excluding the intercept are often referred to as *Sheffé models*, whereas those with the intercept are referred to as *Cox models*. Normally, a full Sheffé model includes all higher order interaction terms, and for this design is given by

$$y = b_1x_1 + b_2x_2 + b_3x_3 + b_{12}x_1x_2 + b_{13}x_1x_3 + b_{23}x_2x_3 + b_{123}x_1x_2x_3$$

As seven experiments have been performed, all the seven terms can be calculated, namely

- three one-factor terms,
- three two-factor interactions and
- one three-factor interaction.

The design matrix is given in Table 2.35, and being a square matrix, the coefficients can easily be determined using the inverse.

The full seven-term Cox model is given by

$$y = a_0 + a_1x_1 + a_2x_2 + a_{11}x_1^2 + a_{22}x_2^2 + a_{12}x_1x_2 + a_{1122}(x_1^2x_2 + x_2^2x_1)$$

Note that only x_1 and x_2 are involved in the model because x_3 is dependent on the other two terms. We could relate the models as $x_1 + x_2 + x_3 = 1$. Note the rather ugly last term that is required if we want to fit a full model. Obviously, for both types of models, we can omit terms to provide some degrees of freedom for the lack-of-fit.

1. Intercept terms are used in
   - **(a)** The full Sheffé model
   - **(b)** The full Cox model
   - **(c)** Both types of model

2.5.2.3 Multi-component Designs

A full simplex centroid design for k components consists of $2^k - 1$ experiments of which there are

- k single blends,
- $k \times (k-1)/2$ binary blends, each component being present in a proportion of ½,
- $k!/[(k-m)!m!]$ blends containing m components (these can be predicted by the binomial theorem or Pascal's triangle), each component being present in a proportion of $1/m$ and
- finally 1 blend consisting of all components, each component being present in a proportion of $1/k$.

Each type of blend yields an equivalent number of interaction terms in the Sheffé model. Hence, for a five-component mixture and three-component blends, there will be $5!/[(5-3)!3!] = 10$ mixtures such as (1/3 1/3 1/3 0 0) containing all possible combinations and 10 terms such as $b_1b_2b_3$.

It is normal to use all possible interaction terms in the mixture model, although this does not leave any degrees of freedom for determining lack-of-fit. Reducing the number of higher order interactions in the model but maintaining

Table 2.35 Design matrix for a three-factor simplex centroid design.

x_1	x_2	x_3	x_1x_2	x_1x_3	x_2x_3	$x_1x_2x_3$
1.000	0.000	0.000	0.000	0.000	0.000	0.000
0.000	1.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	1.000	0.000	0.000	0.000	0.000
0.500	0.500	0.000	0.250	0.000	0.000	0.000
0.500	0.000	0.500	0.000	0.250	0.000	0.000
0.000	0.500	0.500	0.000	0.000	0.250	0.000
0.333	0.333	0.333	0.111	0.111	0.111	0.037

the full design is possible; however, this must be carefully thought because each term can also be re-expressed, in part, as lower order interactions using the Cox model. This will, although, allow the calculation of some measure of confidence in predictions. It is important to recognise that the columns of the mixture design matrix are not orthogonal and can never be because the proportion of each component depends on all others; hence, there will always be some correlation between the factors.

For multi-component mixtures, it is often impracticable to perform a full simplex centroid design; one approach is to simply remove higher order terms, not only from the model but also from the design. A five-component design containing up to second-order terms is presented in Table 2.36. Such designs can be denoted as {*k*,*m*} simplex centroid designs, where *k* is the number of components in the mixture and *m* the highest order interaction. Note that at least binary interactions are required for squared terms (in the Cox model) and so for optimisation.

1. How many possible ternary blends are there for a six-component mixture?
   (a) 20
   (b) 15
   (c) 10
2. How many experiments are required for a {4,2} simplex centroid design?
   (a) 6
   (b) 10
   (c) 11

2.5.3 Simplex Lattice

Another class of designs called *simplex lattice* has been developed and is often preferable to the reduced simplex centroid design when it is required to reduce the number of interaction terms: they span the mixture space more evenly.

A {*k*,*m*} simplex lattice design consists of all possible combinations of 0, 1/*m*, 2/*m* … *m*/*m* or a total of

$$N = (k + m - 1)!/[(k - 1)!m!]$$

experiments where there are *k* factors. A {3,3} simplex lattice design can be set up analogous to the {3,3} simplex centroid design given in Table 2.34. There are

Table 2.36 A {5,2} simplex centroid design.

Comp 1	Comp 2	Comp 3	Comp 4	Comp 5
1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1
1/2	1/2	0	0	0
1/2	0	1/2	0	0
1/2	0	0	1/2	0
1/2	0	0	0	1/2
0	1/2	1/2	0	0
0	1/2	0	1/2	0
0	1/2	0	0	1/2
0	0	1/2	1/2	0
0	0	1/2	0	1/2
0	0	0	1/2	1/2

1. How many experiments are needed for a {5,3} simplex lattice design?
   (a) 21
   (b) 35

2.5.4 Constraints

In chemistry, there are frequent constraints on the proportions of each component. For example, it might be of interest to study the effect of changing the proportion of ingredients in a cake. Sugar will be one ingredient, but there is no point in baking a cake using 100% sugar and 0% of each other ingredients. A more sensible approach is to put a constraint on the amount of sugar, perhaps between 2% and 5%, and look for solutions in this reduced mixture space. A good design will only test blends within the specified regions.

Constrained mixture designs are often quite difficult to set up, but there are four fundamental situations, exemplified in Figure 2.34, each of which requires a different strategy.

- *Only a lower bound* for each component is specified in advance.
  - The first step is to determine whether the proposed lower bounds are feasible. The sum of the lower bounds must be less than 1. For three factors, lower bounds of 0.5, 0.1 and 0.2 are satisfactory, whereas lower bounds of 0.3, 0.4 and 0.5 are not as they add up to more than 1.
  - The next step is to determine new upper bounds. For each component, these are one minus the sum of the lower bounds for all other components. If the lower bounds for three components are 0.5, 0.1 and 0.2, then the upper bound for the first component is $1 - 0.1 - 0.2 = 0.7$; hence, the upper bound of one component plus the lower bounds of the other two must equal 1.

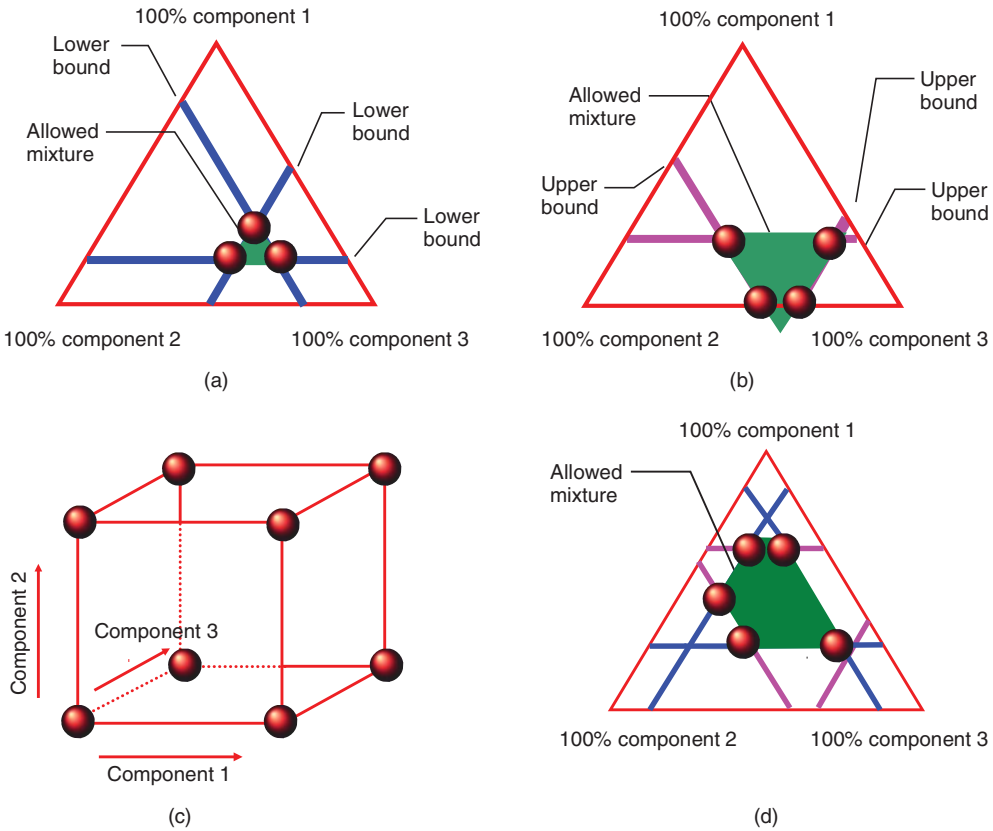


Figure 2.34 Four situations encountered in constrained mixture designs. (a) Lower bounds defined, (b) upper bounds defined, (c) upper and lower bounds defined, fourth factor as filler and (d) upper and lower bounds defined.

Table 2.42 (Continued)

Five vertices				
Lower	0.1	0.3	0	
Upper	0.7	0.6	0.4	
Step 1				
$0.7+0.3+0.0=1.0$				
$0.1+0.6+0.0=0.7$				
$0.1+0.3+0.4=0.8$				
$0.7+0.6+0=1.3$				
$0.7+0.3+0.4=1.4$				
$0.1+0.6+0.4=1.1$				
so $\nu=5$				
Steps 2 and 3 Vertices				
A	0.7	0.3	0.0	
B	0.4	0.6	0.0	
C	0.1	0.6	0.3	
D	0.1	0.5	0.4	
E	0.3	0.3	0.4	
Steps 4 and 5 Design				
1	A	0.7	0.3	0.0
2	Average A&B	0.55	0.45	0.0
3	B	0.4	0.6	0.0
4	Average B&C	0.25	0.6	0.15
5	C	0.1	0.6	0.3
6	Average C&D	0.1	0.55	0.35
7	D	0.1	0.5	0.4
8	Average D&E	0.2	0.4	0.4
9	E	0.3	0.3	0.4
10	Average E&A	0.5	0.3	0.2
11	Centre	0.32	0.46	0.22

2.5.5 Process Variables

Finally, it is useful to briefly mention designs for which there are two types of variables: conventional (often called *process*) variables, such as pH and temperature, and mixture variables, such as solvent composition. A typical experimental design is represented in Figure 2.35, in the case of two process variables and three mixture variables consisting of 28 experiments. Such designs are relatively straightforward to set up, using the principles of this and earlier sections, but care should be taken when calculating a model, which can become very complex. The interested reader is strongly advised to check the detailed literature as it is easy to get very confused when analysing such types of design, although it is important not to get put off, many problems in chemistry involve both types of variables and as there are often interactions, between mixture and process variables (a simple example is that the pH dependence of a reaction depends on solvent composition), such situations can be quite common.

1. A design involving varying solvent composition consisting of methanol and acetone and changing pH and temperature.

   (a) There are one mixture and two process variables.

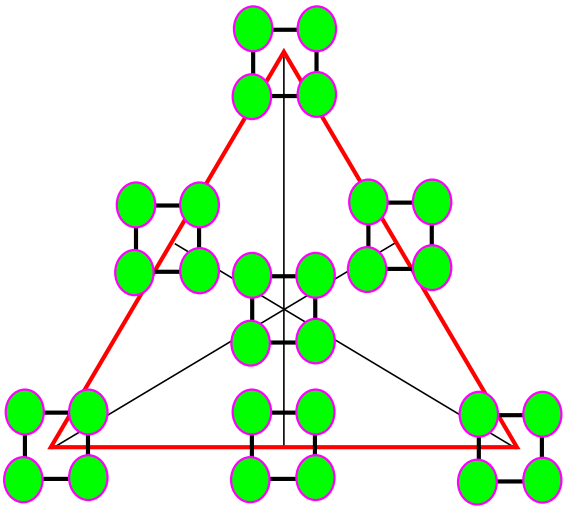


Figure 2.35 Mixture design with process variables.

(b) There are two mixture and two process variables.
(c) All variables are process variables.

2.6 Simplex Optimisation

Experimental designs can be employed for a large variety of purposes, one of the most successful being optimisation. Traditional statistical approaches normally involve forming a mathematical model of a process, and then, either computationally or algebraically, optimising this model to determine the best conditions. There are many applications, however, in which obtaining a mathematical relationship between the response and the factors that influence it are not of primary interest. Is it necessary to model precisely how pH and temperature influence the yield of a reaction? When shimming an NMR machine, is it really important to know the precise relationship between field homogeneity and resolution? In engineering, especially, methods for optimisation have been developed, which do not require a mathematical model of the system. The philosophy is to perform a series of experiments, changing the values of the control parameters, until a desired response is obtained. Statisticians may not like this approach, as it is not normally possible to calculate confidence in the model and the methods may fall down when experiments are highly irreproducible, but in practice, sequential optimisation has been very successfully applied throughout chemistry.

One of the most popular approaches is called *simplex optimisation*. A simplex is the simplest possible object in N dimensional space, for example, a line in one dimension and a triangle in two dimensions, as introduced previously (Figure 2.31). Simplex optimisation implies that a series of experiments are performed on the corners of such a figure. Most simple descriptions are of two-factor designs, where the simplex is a triangle, but, of course, there is no restriction on the number of factors.

2.6.1 Fixed Sized Simplex

The most common, and easiest to understand, method of simplex optimisation is called the *fixed sized simplex*. It is best described as a series of rules.

The main steps are as follows, exemplified by a two-factor experiment.

- Define how many factors are of interest, which we will call k.
- Perform $k + 1$ (=3 in our case) experiments on the vertices of a simplex (or triangle for two factors) in factor space. The conditions for these experiments depend on the step-size. This defines the final 'resolution' of the optimum. The smaller the step-size, the better the optimum can be defined, but the more the experiments necessary. A typical initial simplex might consist of the three experiments, for example
  - pH 3 temperature 30 °C
  - pH 3.01 temperature 31 °C
  - pH 3.02 temperature 30 °C

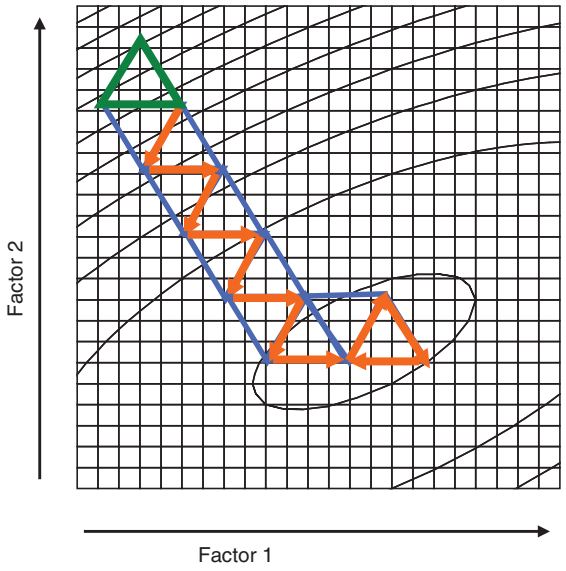


Figure 2.37 Progress of a fixed sized simplex.

1. Six factors are being optimised. How many points are in the simplex?
   (a) 3
   (b) 5
   (c) 6
   (d) 7
2. The step-size of a simplex is defined by the initial conditions.
   (a) True
   (b) False

2.6.2 Elaborations

Many elaborations have been developed over the years. One of the most important is the $k+1$ rule. If a vertex has remained part of the simplex for $k+1$ steps, perform the experiment again. The reason for this is that response surfaces may be noisy, so an unduly optimistic response could have been obtained because of experimental variability. This is especially important when the response surface is flat near the optimum. Another important issue relates to boundary conditions. Sometimes there are physical reasons why a condition cannot cross a boundary, an obvious case being a negative concentration. It is not always easy to deal with such situations, but it is possible to define $\boldsymbol{x}_{new} = \boldsymbol{c} + \boldsymbol{c} - \boldsymbol{x}_2$ rather than $\boldsymbol{x}_{new} = \boldsymbol{c} + \boldsymbol{c} - \boldsymbol{x}_1$ under such circumstances. If the simplex constantly tries to cross a boundary, either the constraints are a little unrealistic and so should be changed or the behaviour near the boundary needs further investigation. Starting a new simplex near the boundary with a small step-size may solve the problem.

1. Significant experimental irreproducibility may cause oscillation around an optimum.
   (a) True
   (b) False

2.6.3 Modified Simplex

A weakness with the standard method for simplex optimisation is a dependence on the initial step-size, which is defined by the initial conditions. For example, in Figure 2.36, we set a very small step-size for both variables; this may be fine if we are quite sure we are near the optimum, otherwise a bigger triangle would reach the optimum quicker. However, the problem is that the bigger step-size may miss the optimum altogether. An alternative method is called the *modified*

1. The modified simplex allows the step-size to both expand and contract.
   (a) False, it can only expand to reach an optimum faster.
   (b) False, it can only contract when close to an optimum to define it better.
   (c) True.

2.6.4 Limitations

In many well-behaved cases, simplex performs well and is quite an efficient approach for optimisation. There are, however, a number of limitations.

- If there is a large amount of experimental error, then the response is not very reproducible. This can cause problems, for example, when searching a fairly flat response surface.
- Sensible initial conditions and scaling (coding) of the factors are essential. This can only come from empirical chemical knowledge.
- If there are serious discontinuities in the response surface, this cannot always be taken into account.
- There is no modelling information. Simplex does not aim to predict unknown responses, produce a mathematical model or test the significance of the model using ANOVA. There is no indication of the size of interactions or related effects.

There is some controversy as to whether simplex methods should genuinely be considered as experimental designs, rather than algorithms for optimisation. Some statisticians often totally ignore this approach, and, indeed, many books and courses of experimental design in chemistry will omit simplex methods altogether, concentrating exclusively on approaches for mathematical modelling of the response surface. However, engineers and programmers have employed simplex and related approaches for optimisation for many years, and these methods have been much used, for example, in spectroscopy and chromatography, thus should be considered by the chemist. As a practical tool where the detailed mathematical relationship between response and underlying variables is not of primary concern, the methods described above are very valuable. They are also easy to implement computationally and to automate and simple to understand.

1. Simplex designs assume there are no interactions.
   (a) True
   (b) False

Problems

2.1 A Two-Factor, Two-Level Design
Section 2.2.3 Section 2.3.1
The following represents the yield of a reaction recorded at two catalyst concentrations and two reaction times

Concentration (mM)	Time (h)	Yield
0.1	2	29.8
0.1	4	22.6
0.2	2	32.6
0.2	4	26.2

1. Obtain the design matrix from the raw data, $\boldsymbol{D}$, containing four coefficients of the form

$$y = b_0 + b_1 x_1 + b_2 x_2 + b_{12} x_1 x_2$$